2026 Wrestling Championships

ML Model Evaluation Report

Post-Tournament Analysis · 4 Machine Learning Models · 3 Conferences · 10 Weight Classes Each (30 Brackets Total)

75.8%
Best Model — All Matches
XGBoost w/ ranks · 344/454 correct

76.6%
Best Model — Regular Matches
XGBoost w/ ranks · 327/427 correct

454
Total Matchups Evaluated
3 Conferences Combined

30
Brackets Predicted
10 Weight Classes × 3 Conferences

COMBINED OVERALL MODEL ACCURACY — ALL 3 CONFERENCES

454 total matchups (all) · 427 regular matchups

Rank	Model	Correct (All Matches)	Accuracy (All Matches)	Correct (Regular Only)	Accuracy (Regular Only)
1	**XGBoost (with ranks) ★**	344 / 454	**75.8%**	327 / 427	**76.6%**
2	XGBoost (no ranks)	335 / 454	**73.8%**	320 / 427	**74.9%**
3	Logistic Regression	329 / 454	**72.5%**	314 / 427	**73.5%**
4	Decision Tree	326 / 454	**71.8%**	313 / 427	**73.3%**

MODEL ACCURACY BY CONFERENCE

Ranked by XGBoost (with ranks) · All Matches vs Regular Only

Conference	Match Type	Logistic Reg.	Decision Tree	XGBoost (no ranks)	★ XGBoost (with ranks)
ACC	All Matches	71.0%	69.2%	72.9%	**75.7%**
	Regular Only	71.3%	69.3%	73.3%	**76.2%**
Big Ten	All Matches	70.8%	70.8%	72.3%	**76.1%**
	Regular Only	72.1%	72.8%	73.7%	**77.4%**
Ivy League	All Matches	79.3%	78.2%	79.3%	**74.7%**
	Regular Only	80.2%	79.1%	80.2%	**74.4%**

** Ivy League exception: XGBoost (with ranks) underperforms vs other models — team_rank_diff carries less signal in the more competitively compressed Ivy League.*

THE PREDICTION MODELS

Four machine learning algorithms — each with different logic and feature sets

LR

MODEL 1

BASELINE MODEL · 1 FEATURE

MODEL 1 — LOGISTIC REGRESSION

Logistic Regression estimates the probability of a binary outcome (win/loss). It finds the best-fit line through data, then applies a **sigmoid function** to convert scores into probabilities between 0 and 1. Simple, interpretable, and serves as a strong baseline — if a wrestler has a significantly higher win rate, this model predicts them to win.

Features:

```
1 feature: win_rate_diff (home win rate minus away win rate)
```

DT

MODEL 2

33 FEATURES

MODEL 2 — DECISION TREE

A Decision Tree builds a flowchart of if/else rules learned from data. It splits on the feature that best separates winners from losers at each node. Naturally handles non-linear relationships — you can trace exactly why the model predicted a given outcome.

Features:

```
33 features: wrestled_before · home_point_diff_rematches · home_pinned_away ·
matches_wrestled_diff · win_rate_diff · bonus_rate_diff · pin_count_diff ·
avg_point_diff_diff · avg_points_scored_diff · std_point_diff_diff · weight_class
encodings (×10) · home academic class encodings (×7) · away academic class encodings
(×7)
```

XGB1

MODEL 3

33 FEATURES

MODEL 3 — XGBOOST v1

XGBoost builds an ensemble of decision trees **sequentially** — each new tree corrects errors of the previous ones. Highly accurate, handles missing data natively, and captures complex non-linear feature interactions that simpler models miss.

Features:

```
33 features (same as Decision Tree above)
```

XGB2

MODEL 4

34 FEATURES

MODEL 4 — XGBOOST v2 ★ BEST MODEL

Same XGBoost algorithm as Model 3, but with one critical additional feature: **team_rank_diff** — the difference in national team rankings between the two wrestlers' programs. This captures coaching quality, depth of daily competition, program resources, and recruiting strength — contextual signals not visible in individual statistics alone.

Features:

```
34 features (all Decision Tree features) + team_rank_diff ← UNIQUE TO THIS MODEL
team_rank_diff reflects that a wrestler from a top-ranked program is likely
battle-tested against elite competition every day in practice.
```

MATCH TYPES: REGULAR vs. SPECIAL vs. EXCLUDED

Understanding what counts in each accuracy figure — and why

This tournament had three categories of match outcomes, each treated differently in the accuracy evaluation. Understanding these distinctions is critical to interpreting the "All Matches" vs. "Regular Matches Only" figures reported throughout this report.

Category	How the Match Ended	Outcome Methods / Details	In 'All Matches'?	In 'Regular Only'?
Regular Match	**One wrestler outcompeted the other through genuine head-to-head action.** The result directly reflects skill, conditioning, and competitive performance on the mat.	Decision · Major Decision · Technical Fall · Fall/Pin · Sudden Victory (OT) · Tiebreaker	✓ Yes	✓ Yes
Special Match	**The match ended due to an Injury Default** — one wrestler was unable to continue because of injury, awarding the win to their opponent by default. No statistical model can predict when a wrestler will suffer an injury mid-match; this outcome is entirely outside the scope of machine learning prediction.	Injury Default (ACC 184 lbs, 3rd-place match) — 1 match total across all conferences.	✓ Yes	✗ No
Excluded (Edge Case)	**Matches involving an unregistered last-minute roster replacement** for whom no predictions were generated. These bouts are neither Regular nor Special — they are simply unpredictable due to the absence of any model data for that wrestler.	Ivy League 285 lbs: Princeton's Sebastian Garibaldi was replaced at the last minute by Anderson Olcott. The 3 matches Olcott competed in are fully excluded from all evaluation.	✗ No	✗ No

■ **Ivy League 285 lbs Edge Case — Last-Minute Roster Change:** Princeton's Sebastian Garibaldi was replaced at the last minute by **Anderson Olcott**. No predictions were generated for Olcott because he was not part of the original tournament roster that the models were trained and evaluated against. This is fundamentally different from a Special match — it is not an injury during competition, but a pre-tournament roster substitution that makes prediction evaluation impossible. All 3 matches Olcott competed in are **entirely excluded** from both "All Matches" and "Regular Only" accuracy figures. The 285 lbs weight class champion prediction (Vincent Mueller) is still evaluated normally based on the championship match itself.

MODEL PERFORMANCE ACROSS CONFERENCES — ALL vs REGULAR MATCHES

Comparison of all 4 models per conference · Blue = All Matches · Green = Regular Only

Key Takeaways

• XGBoost (with ranks) leads in **ACC** (75.7% / 76.2%) and **Big Ten** (76.1% / 77.4%) for both match types.

• **Ivy League exception:** Logistic Regression and XGBoost (no ranks) tie at 79.3% / 80.2%, outperforming XGBoost (with ranks) at 74.7% / 74.4%. Team ranking differences are compressed in the Ivy League.

• Regular-only accuracy is consistently equal to or higher than all-match accuracy, confirming Injury Defaults are unpredictable from statistics.

Big Ten Wrestling Championship 2026

Champion Prediction Results — 10 Weight Classes · 910 Matchups · 260 Evaluated

CHAMPION PREDICTION ACCURACY — BY MODEL

Percentage of weight class champions correctly predicted out of 10 total.

80.0%	70.0%	80.0%	70.0%
Logistic Regression	Decision Tree	XGBoost (no ranks)	XGBoost (with ranks) ★
8/10 champions correct	7/10 champions correct	8/10 champions correct	7/10 champions correct

WEIGHT-BY-WEIGHT CHAMPION PREDICTION BREAKDOWN

How Champion Predictions Work — Round-Robin Methodology

Before the tournament, each model generated predictions for **every possible head-to-head matchup** between all wrestlers in each weight class — a **round-robin format**. For example, if there are 14 wrestlers in a weight class, that produces 91 unique matchups (14 × 13 ÷ 2). Each model independently predicted a winner and win probability for every single one of these matchups.

To determine the predicted **champion**, we identified which wrestler each model predicted to beat the most opponents — i.e., the wrestler who would win the most round-robin matchups according to that model's predictions. This simulates the tournament bracket by asking: "If every wrestler faced every other wrestler, who does this model think would come out on top?"

The table below then compares each model's predicted champion against the **actual tournament champion**. **✓ = the model's predicted champion actually won ✗ = the model's predicted champion did not win**. This gives one correct/incorrect data point per weight class per model (10 total per conference).

Weight	Actual Champion	LogReg	Dec. Tree	XGB1	★ XGB2	Consensus
125 lbs	Luke Lilledahl (Penn State)	Luke Lilledahl ✓	Luke Lilledahl ✓	Luke Lilledahl ✓	Luke Lilledahl ✓	■ Unanimous
133 lbs	Ben Davino (Ohio State)	Marcus Blaze ✗	Marcus Blaze ✗	Marcus Blaze ✗	Marcus Blaze ✗	■ Unanimous
141 lbs	Jesse Mendez (Ohio State)	Jesse Mendez ✓	Jesse Mendez ✓	Jesse Mendez ✓	Jesse Mendez ✓	■ Unanimous
149 lbs	Shayne Van Ness (Penn State)	Shayne Van Ness ✓	Shayne Van Ness ✓	Shayne Van Ness ✓	Shayne Van Ness ✓	■ Unanimous
157 lbs	PJ Duke (Penn State)	Brandon Cannon ✗	Brandon Cannon ✗	Brandon Cannon ✗	Brandon Cannon ✗	■ Unanimous
165 lbs	Mitchell Mesenbrink (Penn State)	Mitchell Mesenbrink ✓	Mitchell Mesenbrink ✓	Mitchell Mesenbrink ✓	Mitchell Mesenbrink ✓	■ Unanimous
174 lbs	Levi Haines (Penn State)	Levi Haines ✓	Levi Haines ✓	Levi Haines ✓	Levi Haines ✓	■ Unanimous
184 lbs	Rocco Welsh (Penn State)	Rocco Welsh ✓	Max McEnelly ✗	Rocco Welsh ✓	Rocco Welsh ✓	■ 3/4 Models
197 lbs	Josh Barr (Penn State)	Josh Barr ✓	Josh Barr ✓	Josh Barr ✓	Josh Barr ✓	■ Unanimous

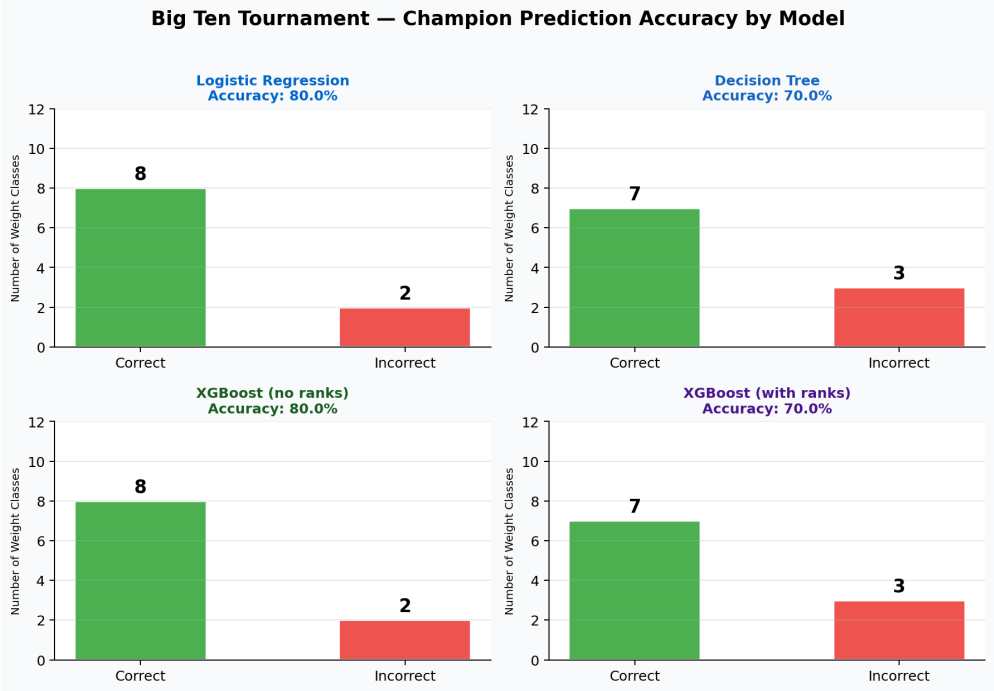
Weight	Actual Champion	LogReg	Dec. Tree	XGB1	★ XGB2	Consensus
285 lbs	Taye Ghadiali (Michigan)	**Taye Ghadiali ✓**	**Taye Ghadiali ✓**	**Taye Ghadiali ✓**	AJ Ferrari ✗	**■ 3/4 Models**

Notes: *133 lbs — all 4 models predicted Marcus Blaze; actual champion was Ben Davino (Ohio State). 157 lbs — all models predicted Brandon Cannon; actual champion was PJ Duke (Penn State). Both were unanimous wrong predictions.*

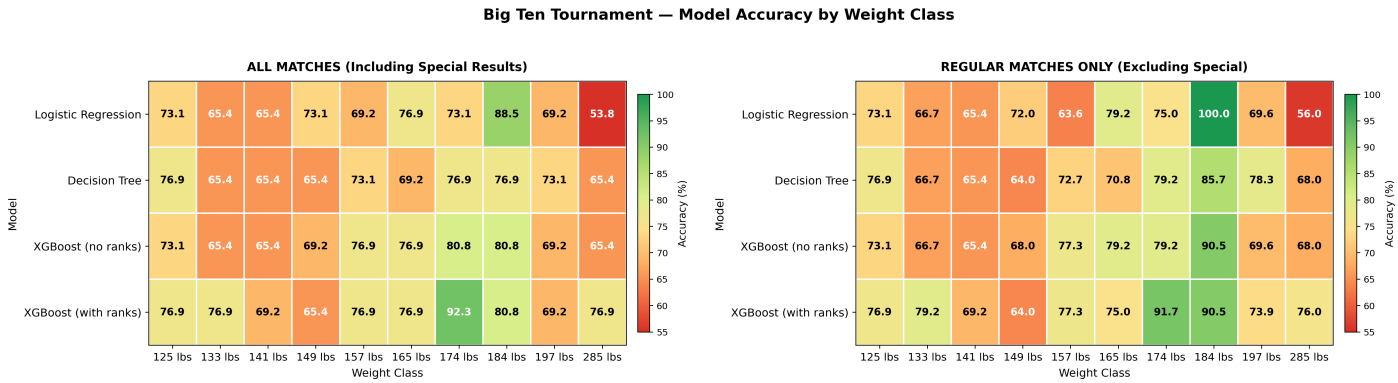
Big Ten — Model Accuracy Visualizations

Left: Champion prediction correct/incorrect bars · Right: Match-level accuracy heatmap

CHAMPION PREDICTION ACCURACY BY MODEL — Each bar shows how many of the 10 weight class champions each model got correct (green) or wrong (red)



MATCH-LEVEL ACCURACY HEATMAP BY WEIGHT CLASS — Each cell = % of individual matchups predicted correctly for that model × weight class combination



Ivy League Wrestling Championship 2026

Champion Prediction Results — 10 Weight Classes · 150 Matchups · 87 All / 86 Regular

CHAMPION PREDICTION ACCURACY — BY MODEL

Percentage of weight class champions correctly predicted out of 10 total.

70.0%	60.0%	70.0%	60.0%
Logistic Regression	Decision Tree	XGBoost (no ranks)	XGBoost (with ranks) ★
7/10 champions correct	6/10 champions correct	7/10 champions correct	6/10 champions correct

WEIGHT-BY-WEIGHT CHAMPION PREDICTION BREAKDOWN

How Champion Predictions Work — Round-Robin Methodology

Before the tournament, each model generated predictions for **every possible head-to-head matchup** between all wrestlers in each weight class — a **round-robin format**. For example, if there are 14 wrestlers in a weight class, that produces 91 unique matchups (14 × 13 ÷ 2). Each model independently predicted a winner and win probability for every single one of these matchups.

To determine the predicted **champion**, we identified which wrestler each model predicted to beat the most opponents — i.e., the wrestler who would win the most round-robin matchups according to that model's predictions. This simulates the tournament bracket by asking: "If every wrestler faced every other wrestler, who does this model think would come out on top?"

The table below then compares each model's predicted champion against the **actual tournament champion**. **✓ = the model's predicted champion actually won ✗ = the model's predicted champion did not win**. This gives one correct/incorrect data point per weight class per model (10 total per conference).

Weight	Actual Champion	LogReg	Dec. Tree	XGB1	★ XGB2	Consensus
125 lbs	Marc-Anthony McGowan (Princeton)	Marc-Anthony McGowan ✓	Marc-Anthony McGowan ✓	Marc-Anthony McGowan ✓	Marc-Anthony McGowan ✓	■ Unanimous
133 lbs	Tyler Ferrera (Cornell)	Evan Mougalian ✗	Evan Mougalian ✗	Evin Gursoy ✗	Evin Gursoy ✗	■■ Split
141 lbs	Vince Cornella (Cornell)	Vince Cornella ✓	CJ Composto ✗	Vince Cornella ✓	Vince Cornella ✓	■ 3/4 Models
149 lbs	Jaxon Joy (Cornell)	Jaxon Joy ✓	Jaxon Joy ✓	Jaxon Joy ✓	Jaxon Joy ✓	■ Unanimous
157 lbs	Jude Swisher (Pennsylvania)	Meyer Shapiro ✗	Meyer Shapiro ✗	Meyer Shapiro ✗	Meyer Shapiro ✗	■ Unanimous
165 lbs	Cesar Alvan (Columbia)	Cesar Alvan ✓	Ty Whalen ✗	Cesar Alvan ✓	Cesar Alvan ✓	■ 3/4 Models
174 lbs	Simon Ruiz (Cornell)	Simon Ruiz ✓	Simon Ruiz ✓	Nick Fine ✗	Nick Fine ✗	■■ Split
184 lbs	Joe Curtis (Columbia)	Joe Curtis ✓	Joe Curtis ✓	Joe Curtis ✓	Joe Curtis ✓	■ Unanimous
197 lbs	Andrew Reall (Brown)	Andrew Reall ✓	Andrew Reall ✓	Andrew Reall ✓	Jack Wehmeyer ✗	■ 3/4 Models
285 lbs	Vincent Mueller (Columbia)	Alex Semenenko ✗	Vincent Mueller ✓	Vincent Mueller ✓	Vincent Mueller ✓	■ 3/4 Models

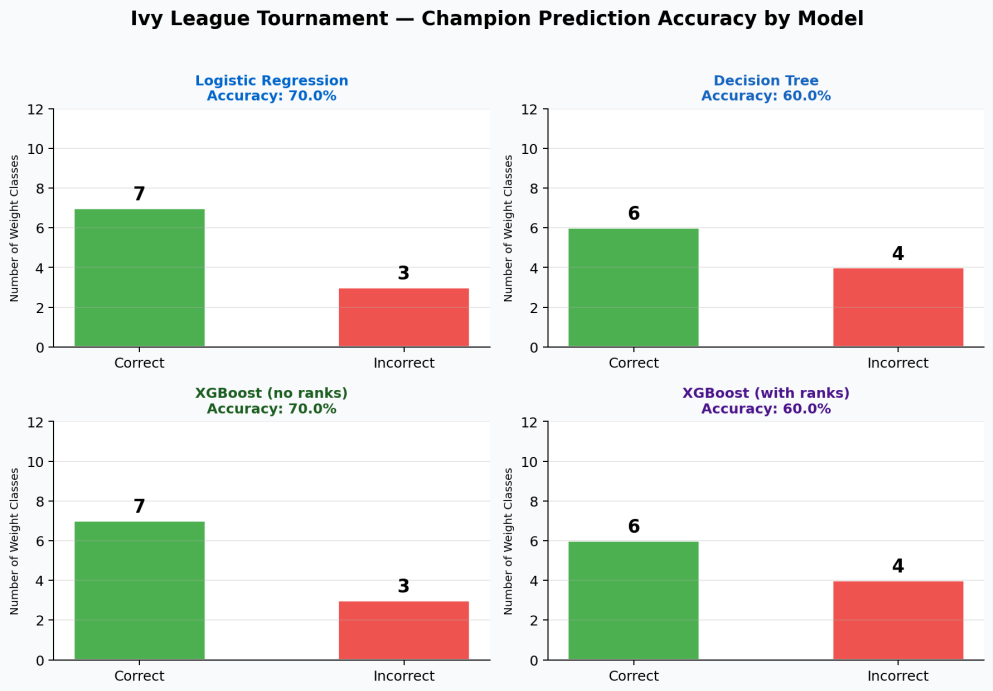
Notes: *133 lbs — Tyler Ferrera (Cornell) won; no model predicted him (LR/DT→Mougalian, XGB1/XGB2→Gursoy). 157 lbs — all models predicted Meyer Shapiro; actual champion was Jude Swisher (Pennsylvania). 174 lbs — LR/DT correctly picked Simon Ruiz; XGB models picked Nick Fine.*

■ **285 lbs — Last-Minute Roster Substitution (Edge Case):** Princeton's Sebastian Garibaldi was replaced at the last minute by **Anderson Olcott**. No predictions were generated for Olcott — he was not part of the roster the models evaluated. This is *not* classified as a Regular or Special match; it is a unique edge case. All 3 matches Olcott competed in are **entirely excluded** from both "All Matches" and "Regular Only" accuracy figures. The 285 lbs *champion prediction* (Vincent Mueller vs. the field) is still evaluated normally — only Olcott's specific bouts are excluded.

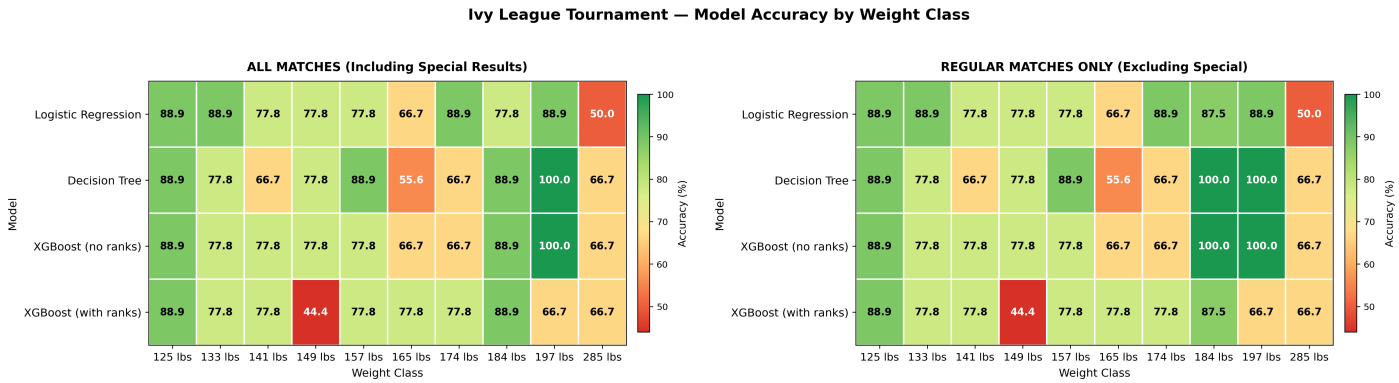
Ivy League — Model Accuracy Visualizations

Left: Champion prediction correct/incorrect bars · Right: Match-level accuracy heatmap

CHAMPION PREDICTION ACCURACY BY MODEL — Each bar shows how many of the 10 weight class champions each model got correct (green) or wrong (red)



MATCH-LEVEL ACCURACY HEATMAP BY WEIGHT CLASS — Each cell = % of individual matchups predicted correctly for that model × weight class combination



ACC Wrestling Championship 2026

Champion Prediction Results — 10 Weight Classes · 210 Matchups · 107 All / 101 Regular

CHAMPION PREDICTION ACCURACY — BY MODEL

Percentage of weight class champions correctly predicted out of 10 total.

40.0%	50.0%	60.0%	60.0%
Logistic Regression	Decision Tree	XGBoost (no ranks)	XGBoost (with ranks) ★
4/10 champions correct	5/10 champions correct	6/10 champions correct	6/10 champions correct

WEIGHT-BY-WEIGHT CHAMPION PREDICTION BREAKDOWN

How Champion Predictions Work — Round-Robin Methodology

Before the tournament, each model generated predictions for **every possible head-to-head matchup** between all wrestlers in each weight class — a **round-robin format**. For example, if there are 14 wrestlers in a weight class, that produces 91 unique matchups (14 × 13 ÷ 2). Each model independently predicted a winner and win probability for every single one of these matchups.

To determine the predicted **champion**, we identified which wrestler each model predicted to beat the most opponents — i.e., the wrestler who would win the most round-robin matchups according to that model's predictions. This simulates the tournament bracket by asking: "If every wrestler faced every other wrestler, who does this model think would come out on top?"

The table below then compares each model's predicted champion against the **actual tournament champion**. **✓ = the model's predicted champion actually won ✗ = the model's predicted champion did not win**. This gives one correct/incorrect data point per weight class per model (10 total per conference).

Weight	Actual Champion	LogReg	Dec. Tree	XGB1	★ XGB2	Consensus
125 lbs	Eddie Ventresca (Virginia Tech)	Eddie Ventresca ✓	Eddie Ventresca ✓	Eddie Ventresca ✓	Eddie Ventresca ✓	■ Unanimous
133 lbs	Aaron Seidel (Virginia Tech)	Tyler Knox ✗	Aaron Seidel ✓	Aaron Seidel ✓	Aaron Seidel ✓	■ 3/4 Models
141 lbs	Jack Consiglio (Stanford)	Gable Porter ✗	Gable Porter ✗	Ryan Jack ✗	Ryan Jack ✗	■■ Split
149 lbs	Collin Gaj (Virginia Tech)	Koy Buesgens ✗	Koy Buesgens ✗	Koy Buesgens ✗	Koy Buesgens ✗	■ Unanimous
157 lbs	Daniel Cardenas (Stanford)	Daniel Cardenas ✓	Daniel Cardenas ✓	Daniel Cardenas ✓	Daniel Cardenas ✓	■ Unanimous
165 lbs	Bryce Hepner (North Carolina)	Will Denny ✗	Will Denny ✗	Will Denny ✗	Will Denny ✗	■ Unanimous
174 lbs	Matthew Singleton (NC State)	Aiden Wallace ✗	Matthew Singleton ✓	Matthew Singleton ✓	Matthew Singleton ✓	■ 3/4 Models
184 lbs	Jaden Bullock (Virginia Tech)	Jake Dailey ✗	Jake Dailey ✗	Jake Dailey ✗	Jaden Bullock ✓	■ 3/4 Models
197 lbs	Sonny Sasso (Virginia Tech)	Sonny Sasso ✓	Mac Stout ✗	Sonny Sasso ✓	Angelo Posada ✗	■■ Split

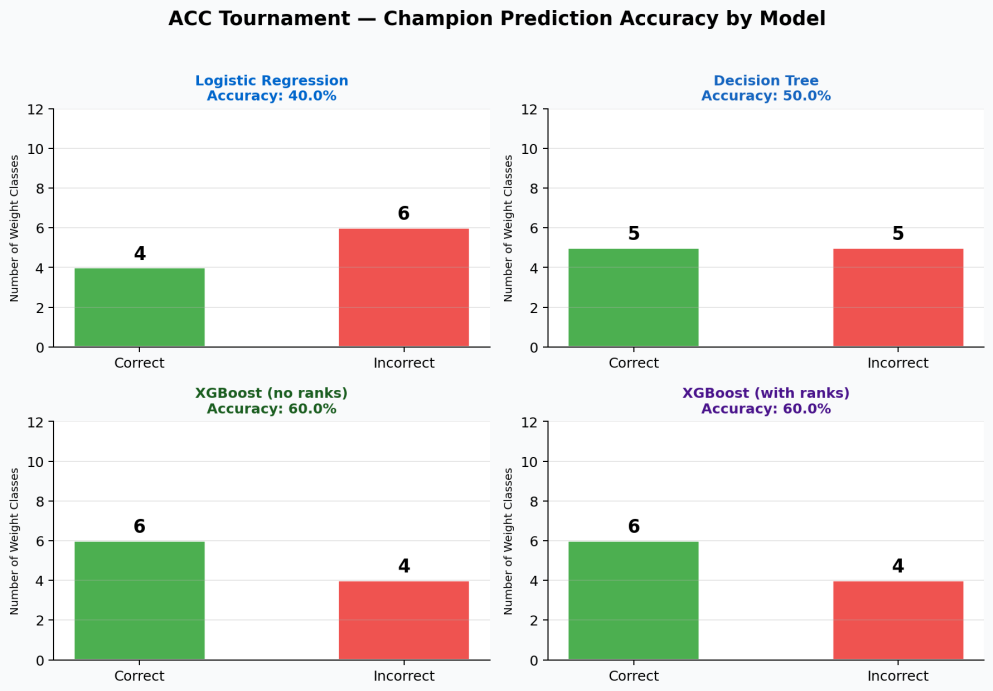
Weight	Actual Champion	LogReg	Dec. Tree	XGB1	★ XGB2	Consensus
285 lbs	Isaac Trumble (NC State)	**Isaac Trumble ✓**	**Isaac Trumble ✓**	**Isaac Trumble ✓**	**Isaac Trumble ✓**	**■ Unanimous**

Notes: *141 lbs — actual champion Jack Consiglio (Stanford) was not predicted by any model. 149 lbs — all 4 models predicted Koy Buesgens; actual champion was Collin Gaj (Virginia Tech). 165 lbs — all models predicted Will Denny; Bryce Hepner (North Carolina) won.*

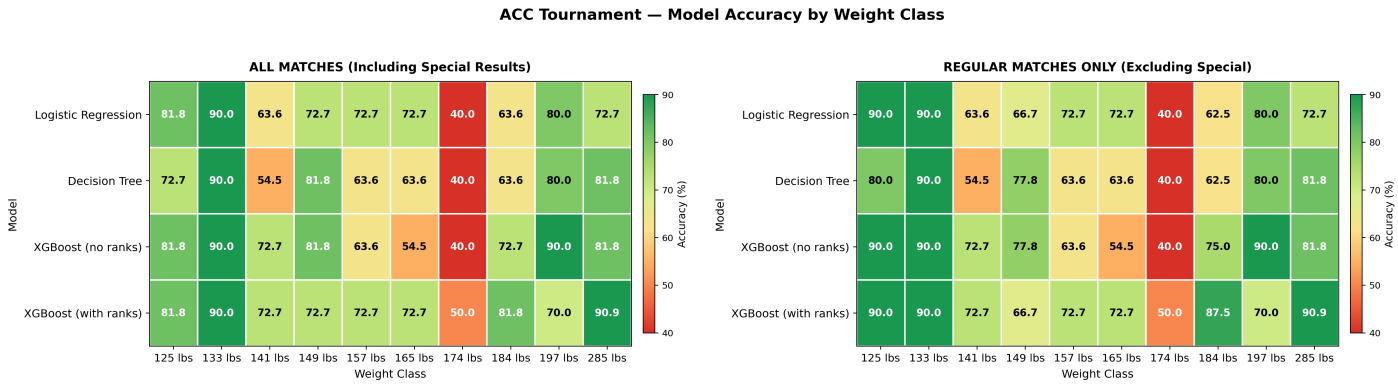
ACC — Model Accuracy Visualizations

Left: Champion prediction correct/incorrect bars · Right: Match-level accuracy heatmap

CHAMPION PREDICTION ACCURACY BY MODEL — Each bar shows how many of the 10 weight class champions each model got correct (green) or wrong (red)



MATCH-LEVEL ACCURACY HEATMAP BY WEIGHT CLASS — Each cell = % of individual matchups predicted correctly for that model × weight class combination



FINAL REPORT SUMMARY

2026 Wrestling Championships — All 3 Conferences

CHAMPION PREDICTION ACCURACY BY CONFERENCE

How often each model correctly predicted the weight class champion — out of 10 per conference

Big Ten

Champion Prediction Accuracy

LR: **80.0%**

DT: **70.0%**

XGB1: **80.0%**

XGB2 ★: **70.0%**

Ivy League

Champion Prediction Accuracy

LR: **70.0%**

DT: **60.0%**

XGB1: **70.0%**

XGB2 ★: **60.0%**

ACC

Champion Prediction Accuracy

LR: **40.0%**

DT: **50.0%**

XGB1: **60.0%**

XGB2 ★: **60.0%**

OVERALL MATCH-LEVEL ACCURACY BY CONFERENCE

Individual matchup predictions — not just champions — across all weight classes

The tables below show **match-level prediction accuracy** — how often each model correctly predicted the winner of *each individual matchup* (not just the champion). This is based on the full round-robin prediction sets across all weight classes.

Conference	Matchups (All)	Matchups (Regular)	LR Acc. (All / Reg)	DT Acc. (All / Reg)	XGB1 Acc. (All / Reg)	★ XGB2 Acc. (All / Reg)
ACC	107	101	71.0% / 71.3%	69.2% / 69.3%	72.9% / 73.3%	**75.7% / 76.2%**
Big Ten	260	240	70.8% / 72.1%	70.8% / 72.8%	72.3% / 73.7%	**76.1% / 77.4%**
Ivy League	87	86	79.3% / 80.2%	78.2% / 79.1%	79.3% / 80.2%	**74.7% / 74.4%**
COMBINED	454	427	72.5% / 73.5%	71.8% / 73.3%	73.8% / 74.9%	**75.8% / 76.6%**

COMBINED OVERALL ACCURACY — ALL CONFERENCES (FINAL)

454 all-match evaluations · 427 regular-match evaluations

Rank	Model	All Matches Correct	All Matches Accuracy	Regular Only Correct	Regular Only Accuracy
1	**XGBoost (with ranks) ★**	344 / 454	**75.8%**	327 / 427	**76.6%**
2	XGBoost (no ranks)	335 / 454	**73.8%**	320 / 427	**74.9%**

3	Logistic Regression	329 / 454	**72.5%**	314 / 427	**73.5%**
4	Decision Tree	326 / 454	**71.8%**	313 / 427	**73.3%**

*★ **XGBoost (with ranks)** is the top overall performer across all 3 conferences combined. Adding **team_rank_diff** provides a meaningful edge in conferences with strong program quality variation (ACC, Big Ten). The Ivy League is the exception — all models perform similarly and team ranking differentials carry less predictive signal.*